ShadeCraft: Learning Time-Dynamic Urban Shadow Patterns with Deep Generative Models

Replicating and Improving Diffusion Model for High-Resolution Shade Simulation in Hot-Arid Cities

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Abstract

Urban heat exposure is a growing public-health concern in hot-arid regions, where limited vegetation and dense built environments intensify surface temperatures. Recent advances in generative modeling have demonstrated promise in simulating complex illumination and shading dynamics at city scale. DeepShade [1], a diffusion-based, text-conditioned framework, offers a notable step forward by integrating satellite imagery, OpenStreetMap (OSM) building geometries, and 3D simulation pipelines to generate time-aware shade maps. However, the generated shade lines often show softened boundaries and minor geometric misalignment, reflecting limitations in the current ControlNet-based conditioning and edge-only structure cues.

In this project, we replicate the DeepShade pipeline and introduce a lightweight architectural improvement to enhance boundary precision. Drawing inspiration from structural refinement modules in ShadowGAN [9] and illumination-aware modeling in Deep Umbra [5], we incorporate a Gradient-Aware Refinement Block (GARB) following the ControlNet backbone. This block fuses multi-scale gradient features derived from both the input edge image and the intermediate latent representation, enabling the model to better preserve high-frequency shadow contours and building outlines.

Our experimental evaluation on the DeepShade Tempe dataset demonstrates that GARB achieves a 27.11% relative improvement in Boundary IoU (B-IoU) and a 5.86% improvement in mean Intersection over Union (mIoU) compared to the baseline, confirming that targeted architectural refinements can meaningfully enhance shade boundary accuracy. These improvements support more reliable downstream urban-heat analyses, heat-adaptive pedestrian routing, and urban-scale shade planning applications.

CCS Concepts

• **Computing methodologies** → **Image generation**; • **Information systems** → *Geographic information systems*; • **Applied computing** → *Environmental sciences.*

Keywords

Urban heat; Shade simulation; Generative models; Diffusion models; Shadow synthesis; Remote sensing; Urban microclimate; Land surface temperature; Tree canopy; Geospatial machine learning

1 Introduction

Urban heat exposure poses a growing challenge in hot-arid cities, where sparse vegetation and dense built forms intensify surface temperatures and reduce pedestrian comfort. Existing urban-climate studies consistently highlight the critical role of accurate shade representation in explaining and predicting local thermal conditions. Remote-sensing-driven analyses of building shadows [8], integrated LST–shade modeling frameworks [6, 7], and field-based evaluations of pedestrian shade exposure [2] all show that even marginal improvements in shade delineation can significantly influence the accuracy of downstream cooling-effect estimates. These findings emphasize that precise shadow boundaries are essential not only for scientific modeling, but also for operational tasks such as heat-resilient routing, city planning, and shade infrastructure design.

Recent advances in generative modeling have opened new possibilities for producing large-scale, temporally flexible shade maps. DeepShade [1], a diffusion-based framework trained on simulated shadow data generated using satellite imagery, OpenStreetMap building geometries, and Blender-based 3D illumination models, introduced a scalable pipeline for city-scale shade synthesis. By leveraging multi-modal conditioning including RGB imagery, Canny edges, and temporal contrastive learning DeepShade demonstrated the ability to generate time-aware shade maps that support applications such as cool-corridor identification and pedestrian heat-aware navigation. However, its reliance on edge-only structural cues and a standard ControlNet backbone creates limitations: shadow boundaries often appear softened, and fine-grained geometric alignment can drift in the presence of complex building outlines or subtle occlusions.

Insights from recent generative shadow modeling research suggest that structural refinement is essential for recovering sharper, more physically plausible shadow edges. ShadowGAN [9] and ARShadowGAN [3] show that adversarial refinement modules, attention blocks, and multi-scale discriminators can enhance the geometric precision of synthetic shadows, particularly in scenes with complex illumination and contextual cues. Meanwhile, Deep Umbra [5], a city-scale conditional GAN for sunlight and shadow accumulation, demonstrates that incorporating gradient features and multi-scalar spatial information dramatically improves generalization across diverse urban morphologies. Collectively, these works indicate that generative models can produce more realistic shading effects when supplemented with hierarchical gradient information rather than relying solely on binary edge maps.

Motivated by these findings, our project builds directly upon the DeepShade pipeline while introducing a lightweight architectural extension designed to enhance boundary fidelity: the Gradient-Aware Refinement Block (GARB). After replicating the original DeepShade data-processing and ControlNet conditioning pipeline, we integrate GARB as a post-ControlNet refinement module. The block fuses gradient features extracted from both the input Canny edge image and the intermediate latent representation of the diffusion model, allowing the network to reinforce high-frequency shadow transitions before denoising unfolds. By injecting multi-scale gradient-aware features, GARB is designed to better preserve building outlines, narrow occlusions, and subtle geometric transitions that typically degrade under standard diffusion sampling.

By retaining DeepShade's core simulation-driven generative pipeline while adding a targeted refinement module rooted in empirical insights from recent shadow-generation literature [3, 5, 9], our system aims to produce shade maps that are both visually sharper and more geometrically faithful. This enhancement supports improved reliability in downstream analyses such as cumulative cooling-effect estimation [7], heat-risk modeling, and pedestrian routing in hot-arid cities domains where small boundary inaccuracies can meaningfully shift environmental interpretations.

2 Related Work

Research on urban shading, thermal exposure, and generative shadow modeling spans multiple domains across remote sensing, computer vision, and environmental simulation. Early studies in urban climatology examined how built form and vegetation modulate land surface temperature (LST). Yu et al. [8] quantified seasonal variations in building-generated shade using Landsat-8 and digital surface models, demonstrating that greater shadow coverage can reduce surface temperature by more than 3 K on impervious surfaces. Similar investigations combining thermal imagery, 3D GIS, and spatial regression reveal strong cooling contributions from both building and tree shade [6, 7], with evidence that the timing and duration of shade exposure play critical roles in determining urban thermal conditions particularly in hot-arid cities such as Tempe, Arizona [7]. Field-based observational studies further highlight the dynamic nature of human thermal exposure; Igoe et al. [2] showed that pedestrian shade profiles measured at sub-second temporal resolution reveal micro-scale variations that static mapping approaches often overlook.

Parallel work in satellite-based LST simulation underscores the importance of geometric and radiative precision in environmental modeling. Morrison et al. [4] demonstrated that sensor view angle, surface complexity, and 3D morphology can alter estimated LST by several Kelvin, emphasizing the need for accurate geometric representation when interpreting or generating shade-related spatial data. Collectively, these studies establish that shading must be modeled with high spatial fidelity to ensure reliable downstream analyses, aligning with the objectives of our refinement-focused enhancement.

In computer vision, generative approaches to shadow synthesis have advanced rapidly. ShadowGAN [9] introduced a conditional adversarial framework capable of synthesizing realistic shadows for virtual objects without explicit scene geometry, leveraging dual discriminators to separately model local shape precision and global illumination coherence. ARShadowGAN [3] extended this paradigm for augmented reality applications through attention-guided adversarial learning, enabling realistic shadow generation under single-light conditions without explicit illumination calibration. These methods illustrate the value of multi-scale features, attention mechanisms, and refinement modules for generating sharper and more consistent shadows design principles that inform the development of gradient-aware refinements in our work.

Large-scale generative modeling of environmental shading was advanced by Deep Umbra [5], which introduced a conditional GAN trained on data from multiple continents to produce high-resolution sunlight and shadow accumulation maps. Deep Umbra demonstrated strong transferability across diverse morphologies and latitudes and highlighted the feasibility of leveraging generative models for global-scale environmental analysis. However, reliance on open datasets (e.g., OSM building heights) and exclusion of vegetation impose limitations relevant to city-scale shade representation, underscoring the need for architectures capable of integrating fine-grained gradient information.

DeepShade [1] represents the first diffusion-based, text conditioned framework for generative shade simulation at urban scale, integrating remote-sensing imagery, OSM building geometries, and 3D simulation pipelines to produce time-aware shadow maps. Its edge-conditioned ControlNet backbone enables scalable citywide inference, but the resulting shade maps often exhibit softened boundaries and minor geometric misalignments due to dependence on binary Canny edges and limited structural refinement. As research has shown that even small improvements in shadow-boundary accuracy can enhance urban heat modeling and pedestrian-route optimization [7], there is a need for architectural extensions that preserve high-frequency spatial detail within generative pipelines.

Our work builds on these foundations by replicating the DeepShade pipeline and introducing a lightweight Gradient-Aware Refinement Block (GARB) designed to better preserve sharp shadow contours and building-based occlusions. Inspired by the structural refinement mechanisms used in ShadowGAN [9], the contextual illumination modeling in ARShadowGAN [3], and the multi-scale gradient integration strategies of Deep Umbra [5], GARB fuses gradient features from both the input edge map and the intermediate latent representation. This approach aligns with insights from prior studies showing that geometric fidelity is essential for accurate thermal and shade modeling [6], while maintaining DeepShade's original diffusion-driven framework.

3 Data

Because our work extends the DeepShade architecture without altering its generative pipeline or target output format, we adopt the same dataset used in the original study [1]. This dataset provides a large, structured collection of simulated shade maps aligned with real-world geographical imagery, enabling us to directly evaluate whether our Gradient-Aware Refinement Block (GARB) improves boundary precision under the same conditions as the baseline model. The dataset is publicly available on Hugging Face as DARL-ASU/DeepShade and is constructed using a formal simulation pipeline designed to ensure geographic diversity, urban form variability, and temporal richness.

3.1 Dataset Creation Pipeline (DeepShade Framework)

Following DeepShade [1], the dataset is generated through a four-stage pipeline that integrates satellite imagery, OpenStreetMap (OSM) building geometries, and Blender-based illumination simulation. This design addresses the challenge that archived or crowd-sourced geographic information can be incomplete or outdated by combining publicly accessible satellite imagery with generalizable modeling tools capable of inferring shade dynamics at large scale.

3.1.1 Metadata Retrieval. The pipeline begins by retrieving metadata required for 3D illumination simulation. OpenStreetMap (OSM) is used as the primary source for urban geometry, providing attributes such as:

- building height and number of floors
- structural footprints and polygonal outlines
- property type and address metadata
- latitude/longitude coordinates

These features form the basis of a 3D representation of the built environment. Although OSM datasets can contain gaps particularly in developing regions, DeepShade mitigates this through large-scale simulation and cross-alignment with satellite imagery.

3.1.2 Simulation Setup. Using the collected metadata, DeepShade generates shade dynamics through large-scale 3D simulation in Blender. The simulation includes:

- a programmable sun-movement engine based on solar declination and day-time angle
- high-fidelity rendering of the Area of Interest (AOI) at Google Maps tile level 13
- physically grounded light projection to generate temporally varying shadows

The sun-movement controller produces a continuous trajectory for any day of the year and at any time, enabling the dataset to capture fine-grained temporal shade variation.

3.1.3 Data Collection. The simulation produces four synchronized data types that serve as multi-modal inputs or supervisory signals for generative modeling:

1. **Shaded Snapshot (x_{shade})**: captures building-side shading, shadow projections, and full structural visibility under sunlight.
2. **Skeleton Snapshot (x_{sk})**: rendered using identical geometry but with sunlight disabled, providing a pure structural reference.
3. **Satellite Image (x_{sat})**: a real satellite tile aligned spatially with the simulated urban scene.
4. **Text Description (T)**: natural language prompts encoding temporal and solar attributes. These prompts are generated from the underlying solar parameters according to $T = f(\theta_{\text{sun}}, t_{\text{day}})$, where θ_{sun} represents solar angle/declination and t_{day} is the timestamp.

Examples include:

- "Solar declination: -20.7°"
- "Angle: 45°"
- "Right now, it is 6:00 PM in a day."

The combination of these modalities (skeleton + shade + satellite + text) enables the model to learn both structural and temporal aspects of shade formation.

3.1.4 Post-Processing and Ground Truth Extraction. To obtain a clean ground-truth shade map, the structural elements and low-intensity artifacts in the shaded snapshot must be removed. DeepShade uses the following formula:

$$x_{\text{gt}} = x_{\text{shade}} - x_{\text{sk}} - I(x_{\text{shade}} \leq \alpha)$$

where:

- x_{gt}: ground-truth shade
- x_{shade}: full shaded snapshot
- x_{sk}: skeleton image
- $I(\cdot)$: indicator function for noise removal
- α: threshold for low-intensity shadow noise

This operation isolates the true cast shadow by subtracting structural content and filtering out non-shadow grey values.

3.2 Dataset Split and Usage

The full DeepShade dataset contains imagery from 34 cities across six continents. For this project, we use the **Tempe subset**, which focuses on the Tempe, Arizona region and provides a controlled environment for evaluating architectural modifications without the computational overhead of training on the full multi-city dataset.

Following the provided train/test configuration files, we use:

- 2,048 training samples (70%)
- 877 testing samples (30%)

Using a single-city subset is appropriate for our work because we focus on architectural refinement rather than cross-city generalization. This controlled setting allows us to isolate the effect of GARB on boundary precision while ensuring a fair comparison against the baseline trained under identical conditions.

3.3 Public Availability

The full DeepShade dataset is publicly released by the DeepShade team and can be accessed at:

https://huggingface.co/datasets/DARL-ASU/DeepShade

4 Methods

To ensure methodological consistency with prior work, we first replicate the core DeepShade pipeline [1], which serves as our baseline model. DeepShade employs a ControlNet-based conditioning mechanism that processes multiple modalities: satellite RGB imagery, Canny edge maps extracted from OSM-derived building footprints, and natural-language temporal prompts encoding solar angle and time of day. These multi-modal inputs guide a text-conditioned diffusion model capable of generating time-aware shade maps that reflect diurnal changes in shadow geometry. Supervised training relies on the skeleton snapshots and ground-truth shade maps available in the DeepShade dataset, enabling the model to learn structure-aware mappings between urban form and corresponding shade patterns.

4.1 Gradient-Aware Refinement Block (GARB)

To address the softened boundaries and geometric inconsistencies observed in DeepShade outputs, we propose the Gradient-Aware Refinement Block (GARB) as a lightweight architectural enhancement inserted between the ControlNet conditioning stage and the diffusion decoder (see Figure 1). The objective of GARB is to reintegrate high-frequency gradient information that is typically diminished as latent features propagate through the diffusion backbone.

GARB operates on multiple feature scales corresponding to the channel dimensions used in ControlNet's U-Net architecture: 320, 640, and 1280 channels. For each scale, GARB consists of three components:

Edge Projection. The input hint image is normalized to zero mean and unit variance, then projected into the feature space through two convolutional layers with SiLU activations.

Refinement Block. The projected edge features are concatenated with the ControlNet features and passed through a three-layer convolutional network. The final layer is **zero-initialized**, ensuring GARB begins training as an identity function.

Gated Residual Connection. The refined features are combined with the original features through a learned sigmoid gate initialized to produce approximately 12% contribution, allowing the model to adaptively control refinement strength during training.

Figure 2 illustrates the internal dataflow of the GARB module.

4.2 Training Configuration

Both models were trained on the Tempe dataset using the DeepShade training pipeline with a learning rate of 1×10^{-4}, AdamW optimizer, and 50 training epochs.

The baseline model was trained on ASU's Sol supercomputer using the original DeepShade training configuration. The GARB-enhanced model was trained on a local workstation with an NVIDIA RTX 5090 GPU (32GB VRAM), using a batch size of 6 with gradient accumulation of 4 steps (effective batch size 24) and gradient clipping at 1.0 for stability. Training the GARB model for 50 epochs required approximately 3 hours.

Both models were evaluated on the local workstation under identical inference conditions using DDIM sampling with 50 steps and a guidance scale of 9.0.

5 Experiments

Our experimental setup evaluates whether the proposed GARB module enhances shade-boundary fidelity relative to the reproduced DeepShade baseline [1]. We compare the original model and the GARB-augmented variant under identical data splits, input modalities, and sampling schedules to ensure a controlled and fair comparison.

5.1 Evaluation Metrics

We adopt the five metrics used in the original DeepShade evaluation:

- **SSIM (Structural Similarity Index)**: Measures perceptual similarity between predicted and ground-truth images on a 0–1 scale, where higher values indicate greater similarity.
- **MSE (Mean Squared Error)**: Average squared pixel error between prediction and ground truth; lower values indicate better accuracy.
- **mIoU (Mean Intersection over Union)**: Measures overlap between predicted and true shade regions; higher values indicate better region accuracy.
- **B-IoU (Boundary IoU)**: IoU computed only along shade boundaries; higher values indicate sharper, better-aligned shade edges.
- **LPIPS (Learned Perceptual Image Patch Similarity)**: Perceptual distance in deep feature space using AlexNet; lower values indicate the generated image "looks" closer to ground truth.

For shade prediction specifically, mIoU and B-IoU are the most task-relevant metrics as they directly measure shadow position and boundary accuracy, which are critical for downstream urban heat analysis applications [7].

6 Results

6.1 Quantitative Results

Table 1 presents the comparative evaluation between the baseline DeepShade model and our GARB-enhanced variant on the Tempe test set.

Table 1: Quantitative comparison between Baseline (DeepShade) and GARB-enhanced models. ↑ indicates higher is better; ↓ indicates lower is better. Best results in bold.

Metric	Baseline	GARB	% Change
SSIM ↑	0.9697	0.9688	-0.09%
MSE ↓	18.17	**17.74**	-2.34%
mIoU ↑	0.2889	**0.3059**	+5.86%
B-IoU ↑	0.0774	**0.0983**	+27.11%
LPIPS ↓	**0.2692**	0.3321	+23.34%

The results demonstrate that GARB achieves meaningful improvements on the task-relevant metrics. B-IoU improves from 0.0774 to 0.0983, a 27.11% relative improvement, indicating substantially sharper and more accurate shade boundaries. mIoU improves from 0.2889 to 0.3059 (5.86% relative improvement), showing better overall shade region prediction. MSE decreases slightly, indicating improved pixel-level accuracy.

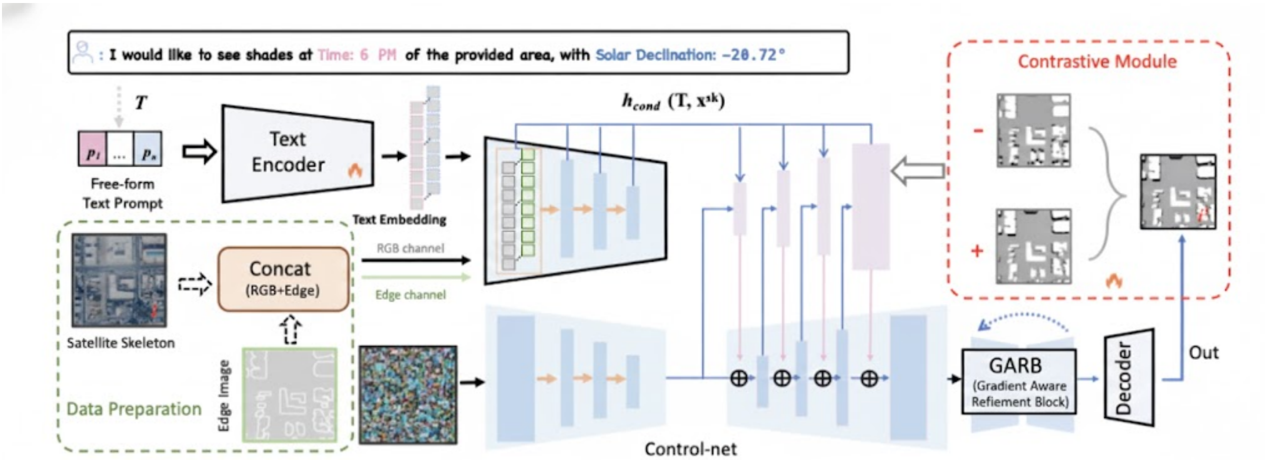

Figure 1: ShadeCraft architecture with GARB integration. The Gradient-Aware Refinement Block is inserted between the ControlNet output and the diffusion decoder, refining multi-scale features before shade map generation. The contrastive module (right) enforces temporal consistency during training. Figure adapted from DeepShade [1].

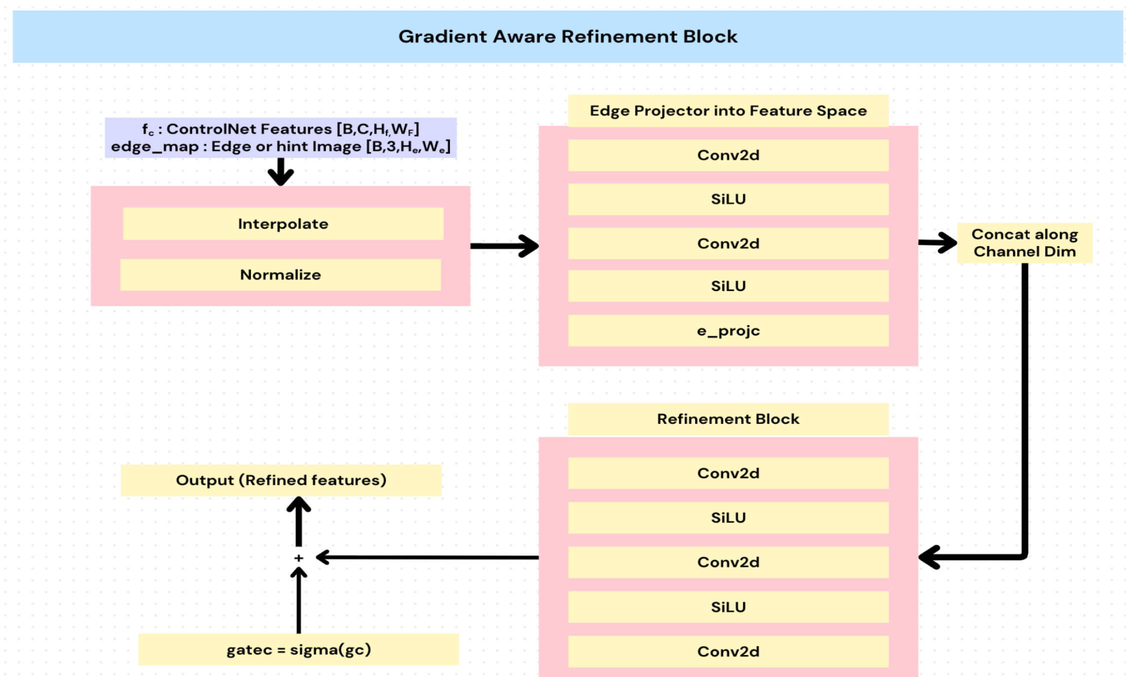

Figure 2: Internal architecture of the Gradient-Aware Refinement Block (GARB). ControlNet features and edge map inputs are processed through an edge projector, concatenated along the channel dimension, and refined through convolutional layers. A learned gating mechanism controls the refinement contribution, which is added residually to the original features.

SSIM remains effectively unchanged (-0.09%), confirming that overall structural similarity is preserved. However, LPIPS increases by 23.34%, indicating reduced perceptual similarity according to deep feature comparison.

6.2 Qualitative Results

Figure 3 illustrates the visual differences between the baseline and GARB models on a representative test sample. The baseline output (left) appears smooth but exhibits soft shade boundaries that do not precisely align with the ground truth (center). The GARB output (right) introduces visible texture noise in non-shadow regions, but

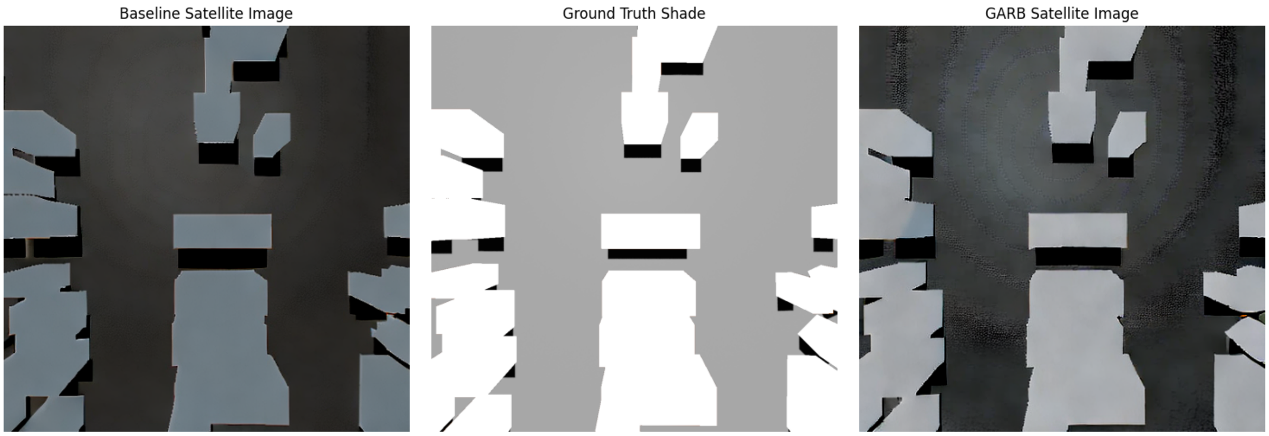

Figure 3: Qualitative comparison on a representative test sample. Left: Baseline output with smooth but soft shade boundaries. Center: Ground truth shade map. Right: GARB output with texture noise but more accurate boundary alignment.

the shade boundaries align more accurately with the ground truth, particularly along building edges.

6.3 Analysis of LPIPS Degradation

The 23.34% increase in LPIPS warrants discussion. LPIPS measures perceptual similarity using features from a neural network trained on general image classification, answering "do these images look similar?" rather than "are the shadows in the correct location?"

Visual inspection reveals that GARB introduces texture noise in non-shadow regions of the generated images. This noise is penalized heavily by LPIPS despite having no impact on shade prediction accuracy. For urban heat modeling applications where shadow position and boundary accuracy are paramount, not visual smoothness, the LPIPS degradation represents an acceptable tradeoff. The improvements in mIoU and B-IoU directly benefit downstream applications such as pedestrian route optimization and cooling-effect estimation [7], where even small boundary inaccuracies can meaningfully shift environmental interpretations.

7 Discussion

7.1 Effectiveness of Multi-Scale Refinement

Our results validate the importance of multi-scale feature refinement for shade boundary accuracy. Early iterations of GARB processed only 320-channel features (the finest scale), which failed to improve upon the baseline. The breakthrough came from extending refinement to all three channel dimensions (320, 640, 1280), allowing GARB to enhance both fine-grained edge details and coarser contextual information simultaneously. This aligns with findings from Deep Umbra [5] and ShadowGAN [9], which emphasize hierarchical feature processing for shadow synthesis.

7.2 Safe Initialization Strategy

The combination of zero-initialized output layers and learned gating proved essential for stable training. Zero initialization ensures GARB starts as an identity function, preventing degradation of pre-trained ControlNet features at the beginning of training. The learned gate, initialized to produce approximately 12% refinement contribution, allows gradients to flow while starting conservatively. During training, the model learns the optimal refinement strength for each scale independently.

7.3 Tradeoff Between Boundary Accuracy and Perceptual Quality

Our results reveal an inherent tension between optimizing for geometric accuracy versus perceptual smoothness. GARB's edge-aware refinement sharpens boundaries but introduces texture artifacts that harm perceptual metrics. This suggests that future work might benefit from auxiliary losses that explicitly penalize texture noise while rewarding boundary accuracy, or from post-processing smoothing applied only to non-shadow regions.

7.4 Limitations

Several limitations should be acknowledged. First, our evaluation uses only the Tempe dataset; generalization to other cities and climates remains untested. Second, the GARB module adds tens of millions of parameters, increasing memory requirements and inference time. Third, the texture noise introduced by GARB, while not affecting shade accuracy, may be undesirable for visualization applications. Finally, our dataset excludes vegetation, limiting applicability to real urban environments where tree shade is significant.

8 Conclusion

We presented ShadeCraft, an extension of the DeepShade framework that incorporates a Gradient-Aware Refinement Block (GARB) to improve shade boundary accuracy in diffusion-based urban shadow simulation. By fusing multi-scale gradient features from the input edge map with intermediate ControlNet representations through a gated residual architecture, GARB achieves a 27.11% relative improvement in Boundary IoU and 5.86% improvement in mean IoU compared to the baseline, while maintaining overall structural similarity.

Our work demonstrates that targeted architectural modifications, specifically, multi-scale edge-aware refinement with safe initialization, can meaningfully enhance task-specific performance in generative models without disrupting the underlying diffusion pipeline. The improvements in shade boundary accuracy directly support more reliable downstream applications including urban heat modeling, pedestrian route optimization, and shade infrastructure planning in hot-arid cities.

8.1 Future Work

Several directions merit future investigation:

- **Model Backbone**: Upgrading from ControlNet Stable Diffusion 2.1 to ControlNet SDXL 1.0 may improve overall generation quality while retaining GARB's boundary refinement benefits.
- **LPIPS Optimization**: Incorporating perceptual smoothness losses or region-aware post-processing could reduce texture artifacts without sacrificing boundary accuracy.
- **Vegetation Integration**: Extending the dataset and model to include tree canopy shadows would improve applicability to real urban environments.
- **Downstream Validation**: Evaluating whether improved B-IoU translates to measurable improvements in pedestrian thermal comfort predictions or route optimization accuracy would validate the practical significance of our contributions.

References

[1] Longchao Da, Xiangrui Liu, Mithun Shivakoti, Thirulogasankar Kutralingam, Yezhou Yang, and Hua Wei. 2025. DeepShade: Enable Shade Simulation by Text-conditioned Image Generation. *arXiv preprint* (2025). doi:10.48550/arxiv.2507.12103

[2] Damien P. Igoe, Nathan J. Downs, Alfio V. Parisi, and Abdurazaq Amar. 2020. Evaluation of shade profiles while walking in urban environments: A case study from inner suburban Sydney, Australia. *Build. Environ.* 177 (2020), 106873. doi:10.1016/j.buildenv.2020.106873

[3] D. Liu, C. Long, H. Zhang, H. Yu, X. Dong, and C. Xiao. 2020. ARShadowGAN: Shadow Generative Adversarial Network for Augmented Reality in Single Light Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8136–8145. doi:10.1109/CVPR42600.2020.00816

[4] William Morrison, Sue Grimmond, and Simone Kotthaus. 2023. Simulating satellite urban land surface temperatures: sensitivity to sensor view angle and assumed landscape complexity. *Remote Sens. Environ.* 293 (2023), 113579. doi:10.1016/j.rse.2023.113579

[5] K. S. Omar, G. Moreira, D. Hodczak, M. Hosseini, N. Colaninno, M. Lage, and F. Miranda. 2024. Deep Umbra: A Generative Approach for Sunlight Access Computation in Urban Spaces. *IEEE Trans. Big Data* (2024). doi:10.1109/TBDATA.2024.3382964 Accepted September 2023.

[6] Y. Park, J. M. Guldmann, and D. Liu. 2021. Impacts of tree and building shades on the urban heat island: Combining remote sensing, 3D digital city and spatial regression approaches. *Comput. Environ. Urban Syst.* 88 (2021), 101655. doi:10.1016/j.compenvurbsys.2021.101655

[7] Yujin Park, Qunshan Zhao, Jean-Michel Guldmann, and Elizabeth A. Wentz. 2023. Quantifying the cumulative cooling effects of 3D building and tree shade with high resolution thermal imagery in a hot arid urban climate. *Landsc. Urban Plan.* 240 (2023), 104874. doi:10.1016/j.landurbplan.2023.104874

[8] Ke Yu, Yunhao Chen, Dandan Wang, Zixuan Chen, Adu Gong, and Jing Li. 2019. Study of the seasonal effect of building shadows on urban land surface temperatures based on remote sensing data. *Remote Sens.* 11, 5 (2019), 497.

[9] S. Zhang, R. Liang, and M. Wang. 2019. ShadowGAN: Shadow synthesis for virtual objects with conditional adversarial networks. *Comput. Vis. Media* 5, 1 (2019), 105–115. doi:10.1007/s41095-019-0136-1